ECE 5252

DIGITAL IMAGE PROCESSING

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DEFINITION

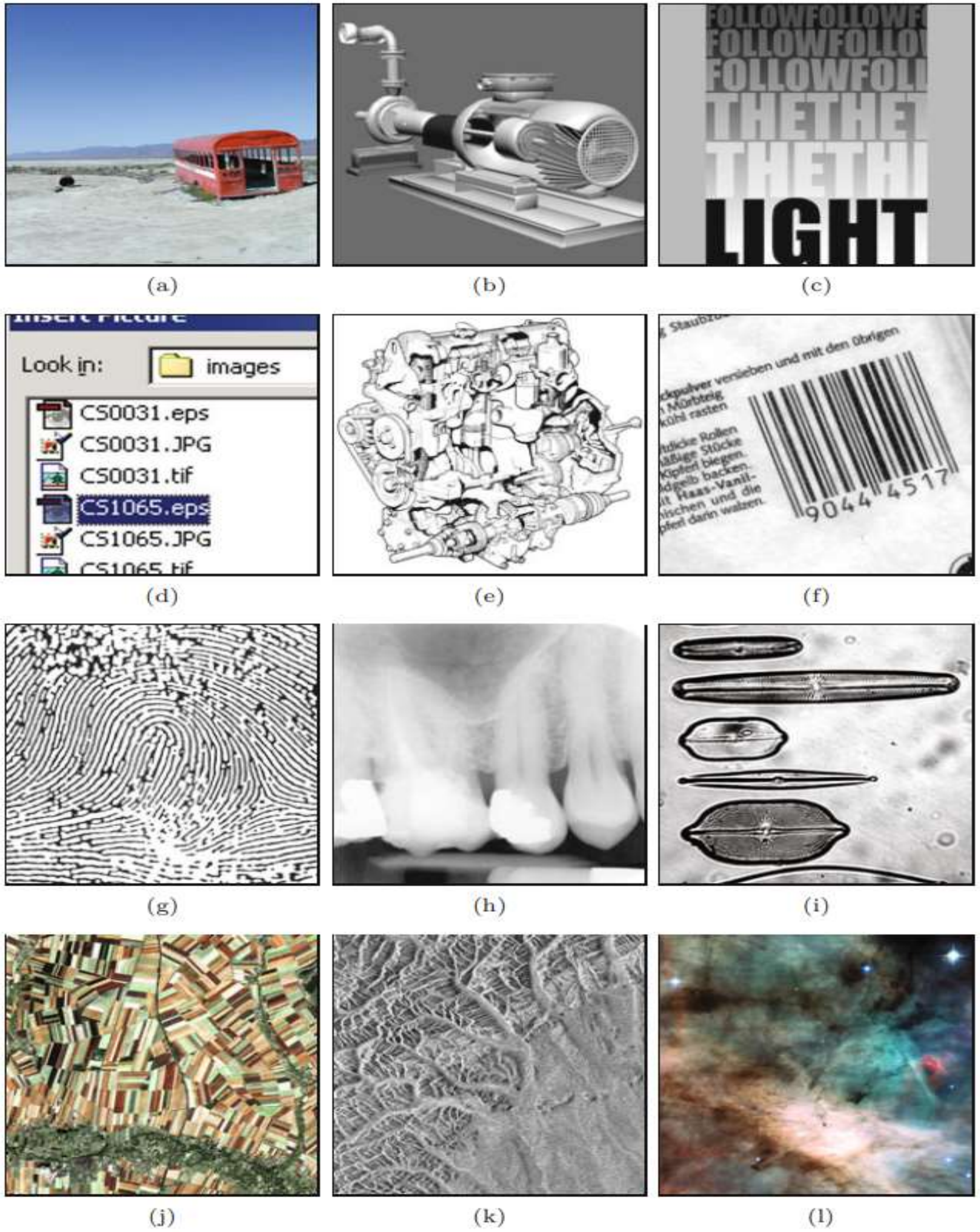
Digital Image

- An **image** is a visual representation of something.

- A **digital image** is an image composed of picture elements, also known as *pixels*, arranged in a regular rectangular grid.

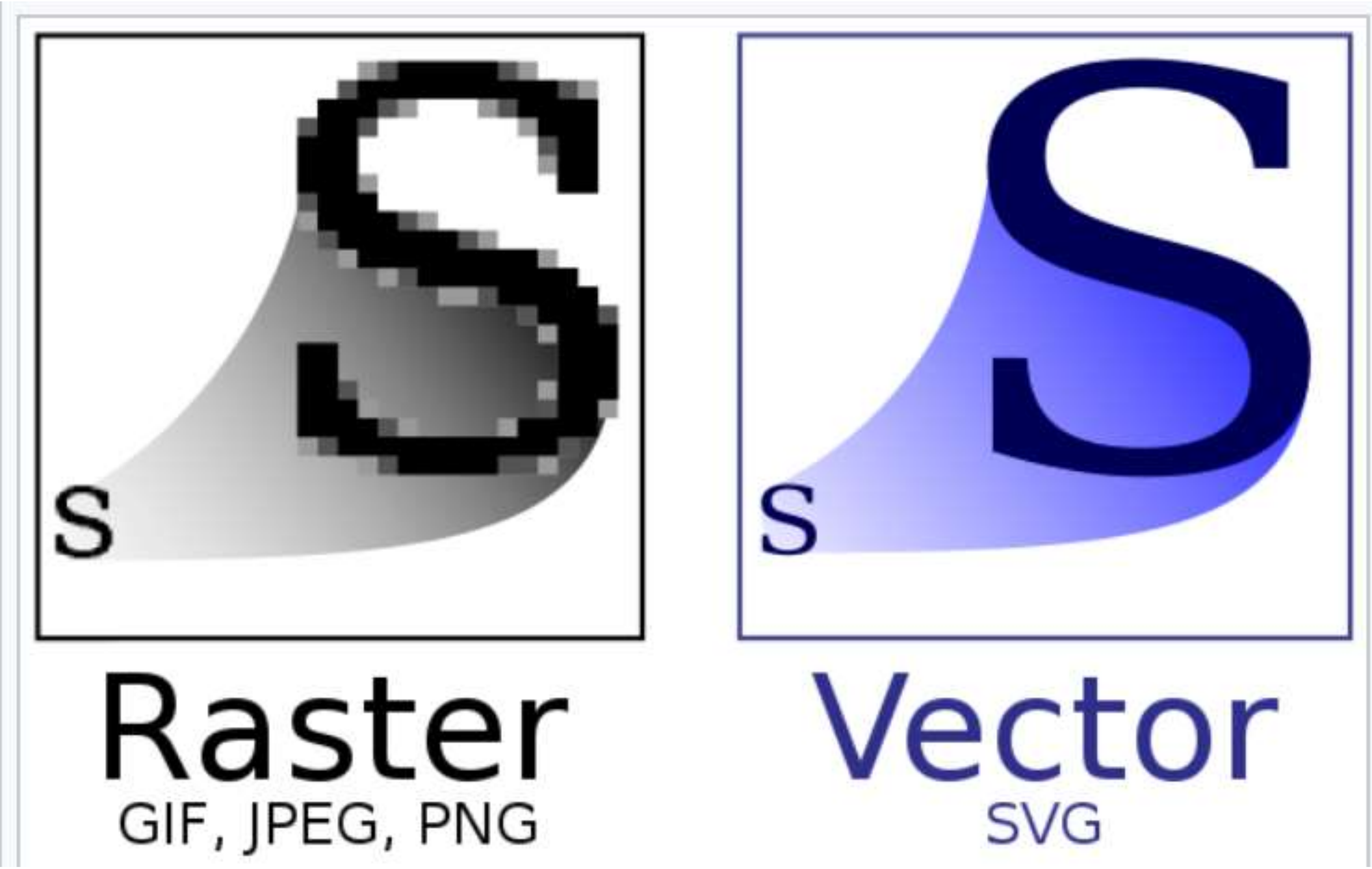
- A **pixel** is the smallest individual element in a raster image.

Digital Images



Examples of digital images. Natural landscape (a), synthetically generated scene (b), poster graphic (c), computer screenshot (d), black and white illustration (e), barcode (f), fingerprint (g), x-ray (h), microscope slide (i), satellite image (j), radar image (k), astronomical object (l).

TYPES OF IMAGE



TYPES OF IMAGE

VECTOR IMAGE

- Vector graphics represent geometric objects using continuous coordinates

- Vector graphics are based on the mathematics of analytic or coordinate geometry

TYPES OF IMAGE

RASTER IMAGE

- have a finite set of digital values, called pixel
- pixel values are arranged in a regular matrix using discrete coordinates
- Pixels hold quantized values that represent the brightness of a given colour at any specific point.
- Pixels are stored in computer memory as a raster image

IMAGE COORDINATES

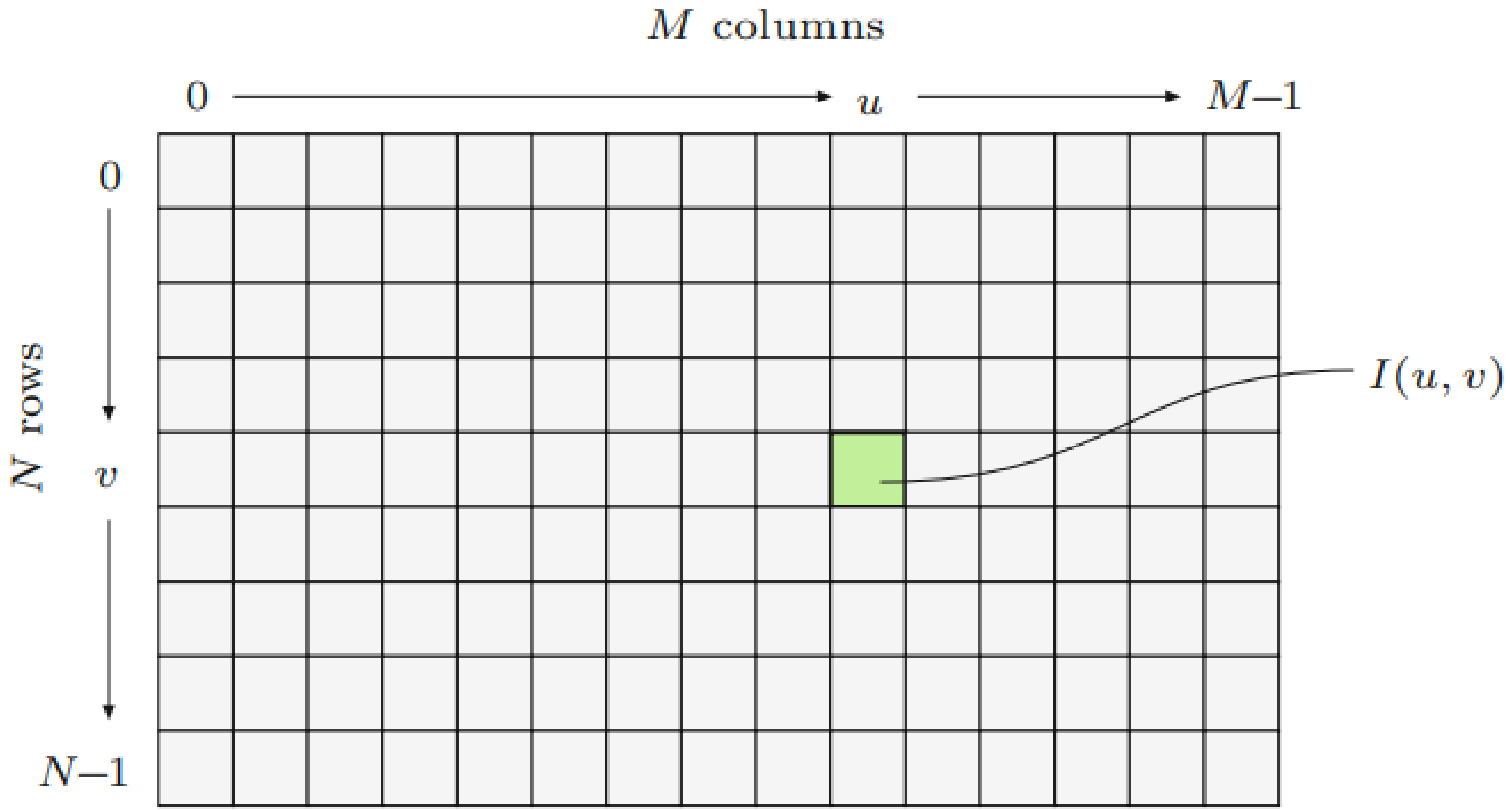


IMAGE COORDINATES

Image coordinates describe the **location of a pixel in an image**

They are written as:

I(u,v) or *I* (column, row)

Origin (0,0) is usually at the **top-left corner**

u increases → to the right

v increases → downward

Example:

If a pixel is at (3,2), it means:

3 pixels to the right, 2 pixels down

IMAGE COORDINATES

The coordinates of the grid on slide 7 are

$$I(9,4)$$

Difference between image coordinates and Matrix

A digital image is stored as a **matrix (2D array)**.

I(row, column) or *I*(v, u)

Concept	Format	Meaning
Image Coordinates	(u, v)	(column, row)
Matrix Indexing	(row, column)	(v, u)

IMAGE COORDINATES

Example

Consider this 3×3 image matrix:

$$\begin{bmatrix} 15 & 20 & 30 \\ 45 & 50 & 60 \\ 70 & 85 & 90 \end{bmatrix}$$

Pixel value **50** is located at:

- Matrix index → M[2,2]
- Image coordinate → (2,2)

But pixel value **60** is:

- Matrix index → M[2,3]
- Image coordinate → (3,2)

Implication

- Image coordinate system → (u, v)
- Matrix indexing → (row, column)
- Relation → (u, v) = (column, row)
- In code → image[v, u]

NOTATIONS AND NOMENCLATURE

Channel – the number of color component in the image that represents the intensity, brightness or density.

Bit depth – the number of bit used to represent one color component, referred to as number of bits per pixel

Pixel values - Pixel values are binary words of length k so that a pixel can represent any of 2^k different values. k = bit depth

NOTATIONS AND NOMENCLATURE

Resolution – is the volume of details an image holds, or the number of image elements (image size) or pixels in an image

Pixel Density

pixels per inch (ppi) – for screen

dots per inch (dpi) – for printing

pixels per kilometer - for satellite images

NOTATIONS AND NOMENCLATURE

Formula:

$$\text{Resolution} = \mathit{Width\ (pixels)} \times \mathit{Height\ (pixels)}$$

Example:

If an image is **1080 × 1920 pixels**, then:

$$1080 \times 1920 = 2{,}073{,}600\ \mathit{pixels}$$

approximately **2.07 megapixels (MP)**.

$$\text{Megapixels} = \frac{\mathit{Total\ Pixels}}{1{,}000{,}000}$$

NOTATIONS AND NOMENCLATURE

Formula:

$$\text{PPI} = \frac{\textit{Number of pixels}}{\textit{Size in inches}}$$

Example:

If a 1080-pixel wide image is printed 6 inches wide:

$$\text{PPI} = \frac{1080}{6} = \textit{180 PPI}$$

Higher PPI = sharper image.

NOTATIONS AND NOMENCLATURE

Image Size – is determined directly from the *width M* (number of columns) and the *height N* (number of rows) of the image matrix. e.g 750 x 500.

Image size can mean 3 things

- Pixel Dimensions (Digital Size) i.e. 750 x 500

  This is sometimes casually called "size," but it is technically the resolution dimensions

- Physical size (Print Size), i.e. 6 inches × 4 inches

  This depends on **PPI (pixels per inch)**.

- File Size (Storage size), i.e. 2MB, 500KB

  This depends on: Resolution, Compression (JPEG, PNG), Color depth

KIND OF DIGITAL IMAGE

GRAYSCALE IMAGE

Binary image: document, illustration, fax

Universal: photo, scan, print

High quality: photo, scan, print

Professional: photo, scan, print

Highest quality: medicine, astronomy

COLOR IMAGE

RGB, universal: photo, scan, print

RGB, high quality: photo, scan, print

RGB, professional: photo, scan, print

CMYK, digital prepress

KIND OF DIGITAL IMAGE

BINARY IMAGE

Binary images are a special type of intensity image where pixels can only take on one of two values, black or white, consisting of a single channel.

These values are typically encoded using a single bit (0/1) per pixel.

Binary images are often used for representing line graphics, archiving documents, encoding fax transmissions, and of course in electronic printing

KIND OF DIGITAL IMAGE

GRAYSCALE IMAGE

The image data in a grayscale image consist of a single channel

Pixel number range from 0,..., 2^k-1.

A typical grayscale image uses k = 8 bits (1 byte) per pixel

Intensity values in the range 0,..., 255, where the value 0 represents the minimum brightness (black) and 255 the maximum brightness (white).

KIND OF IMAGE

COLOR IMAGE

Colour images are based on the primary colours red, green, and blue (RGB).

Colour images use 8 bits per colour component.

In these color images, each pixel requires $3 \times 8 = 24$ bits to encode all three components.

The range of each colour component is [0, 255].

CHARACTERISTICS OF KINDS OF IMAGE

Grayscale (Intensity Images):

Chan.	*Bits/Pix.*	*Range*	*Use*
1	1	$[0,1]$	Binary image: document, illustration, fax
1	8	$[0,255]$	Universal: photo, scan, print
1	12	$[0,4095]$	High quality: photo, scan, print
1	14	$[0,16383]$	Professional: photo, scan, print
1	16	$[0,65535]$	Highest quality: medicine, astronomy

Color Images:

Chan.	*Bits/Pix.*	*Range*	*Use*
3	24	$[0,255]^3$	RGB, universal: photo, scan, print
3	36	$[0,4095]^3$	RGB, high quality: photo, scan, print
3	42	$[0,16383]^3$	RGB, professional: photo, scan, print
4	32	$[0,255]^4$	CMYK, digital prepress

DIGITAL IMAGE PROCESSING

- The manipulation of a digital image using a computer.

- Often used interchangeably with image editing but quite broader than it

DIGITAL IMAGE PROCESSING Cont’d

Digital Image Editing (DIE)

- DIE or Digital Imaging is the manipulation of digital images using an existing software application such as Adobe Photoshop or Corel Paint.

Digital Image Processing (DIP)

- is the conception, design, development, and enhancement of digital imaging programs.

DIGITAL IMAGE PROCESSING Cont’d

Computer Graphics

- concentrates on the synthesis of digital images from geometrical descriptions such as three-dimensional (3D) object model.

HISTORY

PAST

- Digital image processing (i.e. using a computer to manipulate a digital image) was performed by only a relatively small group of specialists

- The specialists who had access to expensive equipment.

- Digital image processing has its roots in the academic realm (since the combination of specialists and equipment are found only in research lab)

HISTORY Cont’d

PAST

- Personal computers were not powerful enough to load a single image into main memory from a typical digital camera

- Digitizing a photo and saving it to a file on a computer was a time-consuming task.

- It was impossible for amateurs to manipulate digital images and videos.

HISTORY Cont'd

PRESENT

- There are powerful computers on every desktop and laptop
- There is ubiquitous access to some devices for digital image acquisition, i.e., cell phone camera, digital camera, or scanner
- Powerful hardware and software packages support image processing
- Amateur works productively with digital images with only a basic understanding of the underlying mechanics; analogous to car driving and a combustion engine

CHALLENGES

- Professionals must be more familiar with digital image processing
- Should be able to knowledgeably manipulate images and related digital media
- Software engineers and computer scientists are confronted with developing programs, databases, and related systems that must correctly deal with digital images.

BENEFITS

- Enables easy access of images and word documents.
- Allows the electronic transmission of images to third-party providers, referring dentists, consultants, and insurance carriers via a modem
- Environmentally friendly since it does not require chemical processing

BENEFITS Cont'd

- Frequently used to help document and record historical, scientific and personal life events

- Digital imaging reduces the need for physical contact with original images

APPLICATION OF DIP

MEDICINE

- Gamma-ray imaging, PET scan, X-Ray Imaging, Medical CT scan,
- UV imaging

1ST ASSIGNMENT

DETERMINE OTHER APPLICATION AREAS OF DIP AND GIVE A BRIEF NARRATIVES OF THE SPECIFIC DIVISIONS.

DIGITAL IMAGE AQUISITION

- Is the process by which a scene becomes a digital image, varied and complicated

DIGITAL IMAGE ACQUISITION DEVICES

- The imaging acquisition devices are composed of the devices of imaging modalities and acquisition gateway computers.

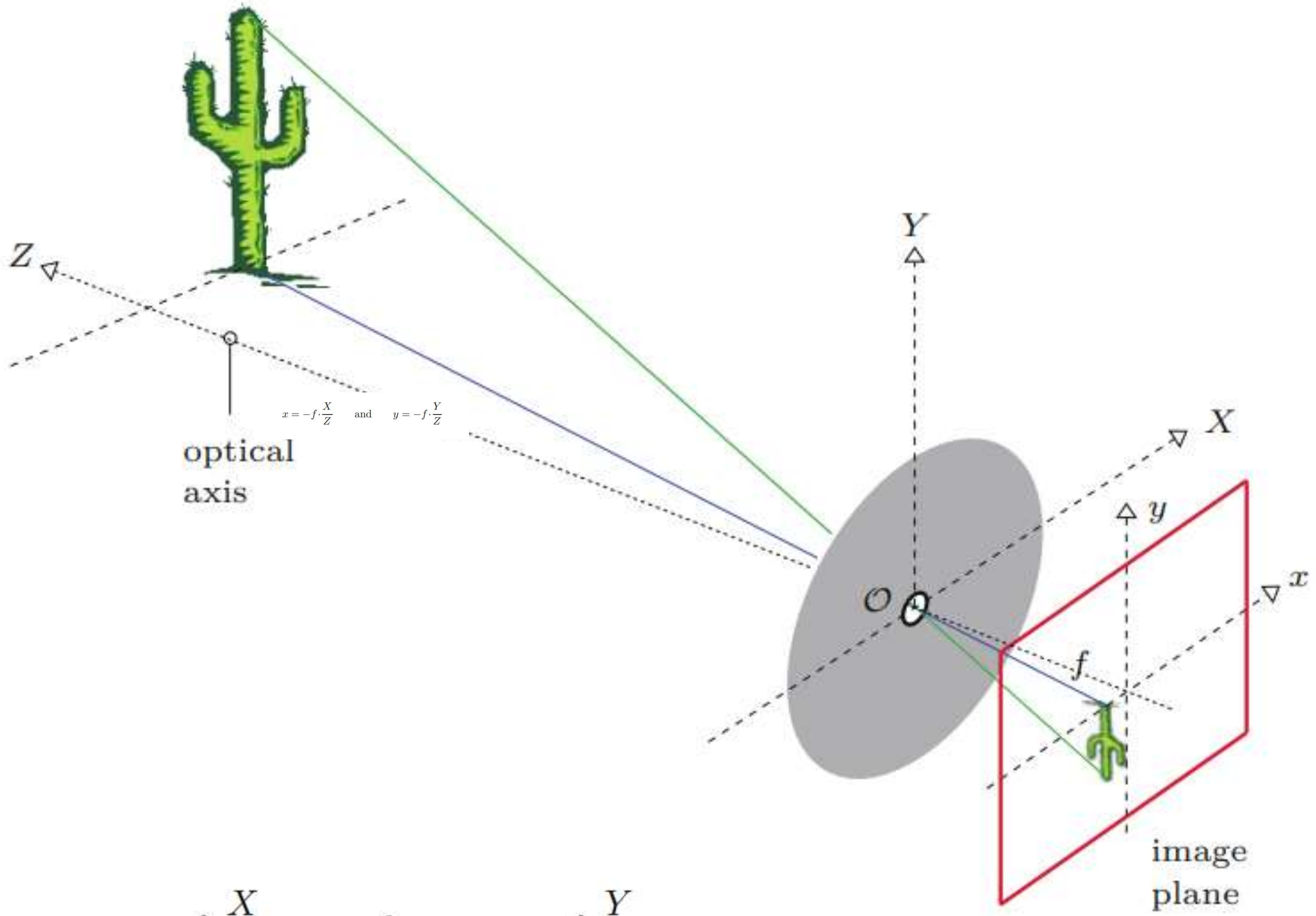
DIGITAL IMAGE ACQUISITION DEVICES Cont’d

Examples

- Cameras
- Fingerprint Scanners
- Iris Scanners
- Magnetic resonance imaging machines
- Computed tomography machines
- Radiography Machine,
- Echocardiogram Machines

THE PINHOLE CAMERA MODEL

Geometry of the pinhole camera. The pinhole opening serves as the origin ($\mathcal{O}$) of the 3D coordinate system (X, Y, Z) for the objects in the scene. The optical axis, which runs through the opening, is the Z axis of this coordinate system. A separate 2D coordinate system (x, y) describes the projection points on the image plane. The distance f ("focal length") between the opening and the image plane determines the scale of the projection.



$$x = -f \cdot \frac{X}{Z} \qquad \text{and} \qquad y = -f \cdot \frac{Y}{Z}$$

THE PINHOLE CAMERA MODEL Cont'd

- For a fixed image, a small f (i.e., short focal length) results in a small image and a large viewing angle, just as occurs when a wide-angle lens is used.

- while increasing the "focal length" f results in a larger image and a smaller viewing angle.

THE PINHOLE CAMERA MODEL Cont'd

1 Important properties of this theoretical model

- are that straight lines in 3D space always appear straight in 2D projections and
- that circles appear as ellipses

IMAGE CONVERSION TO DIGITAL

Characteristics of projected image

- Two-dimensional (2D),
- Time-dependent,
- Continuous distribution of light energy.

Conversion of this image into a digital image on the computer, requires three main steps

IMAGE CONVERSION STEPS

- Step 1: Spatial sampling
- Step 2: Temporal sampling
- Step 3: Quantization of pixel values

Step 1: Spatial sampling

- is the conversion of the continuous signal to its discrete representation
- depends on the geometry of the sensor elements of the acquisition device (e.g., a digital or video camera).

IMAGE CONVERSION STEPS Cont'd

Step 2: Temporal sampling

- Measurement at regular intervals the amount of light incident on each individual sensor element.

- The CCD in a digital camera does this by triggering the charging process and then measuring the amount of electrical charge that has built up during the specified amount of time that the CCD was illuminated.

IMAGE CONVERSION STEPS Cont'd

Step 3: Quantization of pixel values

- This is the scaling of pixel values to an integer scale (e.g., $256 = 2^8$ or $4096 = 2^{12}$).

- Conversion is carried out using an analog to digital converter, which is typically embedded directly in the sensor electronics